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GUIDELINE FOR SELECTION OF REFERENCE PRODUCTS FOR APPLICATIONS OF GENERIC MEDICINES

SUMMARY

This document provides guidance to applicants who seek to apply for new registration of generics supported by Bioequivalence studies. The South African Health Products Regulatory Authority (SAHPRA) can request additional information to establish if the health products meet the safety, quality and performance standards. While other approaches can be used, the processes should be proven scientifically and technically. SAHPRA is committed to ensuring that all registered health products meet the safety, quality and performance standards. Applicants must adhere to the administrative requirements to avoid delays in the processing and evaluation of applications.

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Glossary

Abbreviation/ Term	Meaning
Comparator product	The comparator product, also referred to as the reference product, is a pharmaceutical product with which the multi-source product is intended to be interchangeable in clinical practice. Comparator products will normally be the innovator products for which efficacy, safety, and quality have been established.
Biobatch	The batch used to establish bioequivalence or similarity to the comparator product as determined in bioequivalence or biowaiver studies, respectively.
Extension	Additional products added to the product range are sometimes referred to as 'Line Extensions'. This may include addition of an additional Strengths, Dosage Forms or New Routes of Administration.
BMR	Batch Manufacturing Record
Innovator product	The first registered formulation of a new product, generally a patented medicine based on original scientific documentation of safety, efficacy, and pharmaceutical quality.
New chemical entity	An active pharmaceutical ingredient that has never been used in a marketed finished dosage form in South Africa.
Generic medicine/ Multisource product	Pharmaceutically equivalent or pharmaceutically alternative products that may or may not be therapeutically equivalent. Multisource pharmaceutical products that are therapeutically equivalent are interchangeable.
Market leader	In the absence of an RSA innovator, a market leader is described as the company selling the largest quantity of a particular product.
Old medicine	Unregistered medicines that are on the South African market before registration became mandatory with the promulgation of The Medicines and Related Substances Control Act, 1965 (Act 101 of 1965). These medicines were never formally evaluated by SAHPRA or the former MCC for safety, efficacy, and quality.
PSD	Particle size distribution.
Quality control media	In dissolution studies, QC (quality control) media refers to the specific formulation of the dissolution medium used to assess the release of a drug substance from a dosage form.
Q1/Q2	Generic drugs with certain routes of administration are generally required to contain "the same inactive ingredients in the same concentration as the reference listed drug (RLD)," this is referred to as Q1/Q2 (Qualitative (Q1) and quantitative (Q2)) sameness.

1. INTRODUCTION

This guideline provides appropriate *in vivo* and *in vitro* requirements to assure interchangeability of the multi-source product. Multisource pharmaceutical products are required to meet safety, quality, and performance standards as those applicable to the innovator's product. In addition, reasonable assurance should be provided to ensure that the multisource product is therapeutically equivalent and interchangeable with the product they are compared with.

This guideline should be read in conjunction with the ICH M9, M13, and the Quality and Bioequivalence guidelines, and other applicable guidance.

1.1 Purpose

To register a new generic medicine in South Africa, the applicant must demonstrate bioequivalence against the South African reference product by providing the appropriate *in vivo* and *in vitro* studies as detailed in the quality and bioequivalence guideline. This guideline provides clarity on the selection of a reference product.

A bioequivalence study using an overseas reference product may be acceptable, provided the applicant can demonstrate similarity between the South African and overseas reference products through comparative dissolution.

In line with the adoption of the ICH M13 Guideline, this guideline guides the selection of the reference product for conducting bioequivalence studies.

1.2 Scope

- The choice of comparator product(s) to be used in bioequivalence/ biowaiver applications for generic products for registration.
- The choice of a comparator product for comparative dissolution studies. This also includes the reference products used in comparative dissolution, where the requirements for comparing the foreign reference product with the South African innovator must be met.
- The choice of comparator products for non-solids and semi-solids.

This guideline does not cover the use of products containing chemical entities or active moieties that are not registered in South Africa as reference products to support the safety and efficacy of generic applications, for further information on these products, refer to the Clinical Guideline (SAHPGL-CEM-01).

2. LEGAL PROVISION

Section 15(1)-(3) of the Act states: (1) Every application for the registration of a medicine, medical device or IVD shall be submitted to the Chief Executive Officer in the prescribed form and shall be accompanied by-

(a) the prescribed particulars;

(b) samples of the relevant medicines;

(c) where practicable, samples of medical devices or IVDs; and

(d) the prescribed registration fees.

(2) As soon as possible after receipt by the Chief Executive Officer of an application contemplated in subsection (1), he or she shall inform the applicant in writing that the application is being considered.

(3) (a) If after consideration of any such application and after any investigation or enquiry which it may consider necessary the Authority is satisfied that the medicine, medical device or IVD in question-

(i) is suitable for the purpose for which it is intended;

(ii) complies with the prescribed requirements; and

(iii) is safe, efficacious and of good quality and, in the case of a medical device and IVD, performs as intended, the Authority shall issue the applicant with a certificate of registration to that effect.

3. SELECTION OF A REFERENCE PRODUCT FOR BE STUDIES AND BIOWAIVERS

3.1 Bioequivalence studies

- The choice of the reference product identified by the applicant for which bioequivalence has been demonstrated by appropriate bioavailability studies should be justified in Module 3.2.R.1.1.10.
- Test product in an application for a generic product are normally compared with the corresponding dosage form of a reference product, if available on the market.
- In an application for extension of a medicinal product which has been initially registered with MCC before its changed its name to SAHPRA and when there are several dosage forms of this medicinal product on the market, it is recommended that the dosage form used for the initial

approval of the concerned medicinal product (and which was used in clinical efficacy and safety studies) is used as the reference product, if available on the market. Extensions are handled as new applications refer to guidelines for submission of extension applications (SAHPGL-CEM-PRE-03)

- If a different chemical form is used, it must be confirmed that the safety/efficacy profile is not altered (3.2.R.1.1.11). The confirmation may be documented with bibliographical evidence. If well known (e.g. hydrochloride, maleate, nitrate, stearate), reference to a pharmacopoeia accepted by SAHPRA may be acceptable. Applications with different chemical forms refer to guideline for submission of extension applications (SAHPGL-CEM-PRE-03)
- Product strengths not available in South Africa may be applied for and/ or used in biostudies, provided that the dose range is approved/ registered in South Africa.

3.1.1 Reference Products (comparators)

The reference product should be an innovator product registered by SAHPRA and should preferably be procured in South Africa. If more than one such product is available, the market leader should be used as a reference and consult SAHPRA for guidance. The applicant should submit evidence to substantiate the market leadership claim.

The following options for selection of the reference product are listed in order of preference:

- i) the innovator product registered and procured in South Africa; or
- ii) the innovator product, registered in South Africa, for which a marketing authorisation has been granted by one of the health authorities specified in **Appendix I**. The reference product should be purchased from that market; or
- iii) a product from the latest edition of the WHO List of international comparator products, provided the product is sourced from the market with which SAHPRA aligns itself.

in the case that no innovator product can be obtained/ sourced – within the context of (i)–(iii) above, the choice of the reference product must be made carefully and must be comprehensively justified by the applicant in consultation with SAHPRA (Pre-submission meeting). If the applicant is not able to apply any of the principles outlined in this guideline, including the scenarios/ decision tree, the authority should be consulted before submission of an application. Consultation before purchase of the comparator product is strongly

recommended. Refer to general information guideline (SAHPGL-HPA-07)

NOTE: Comparative dissolution between the foreign reference and the RSA innovator is required for options 3.1.1 ii; iii; above.

Purchase, shipment and storage evidence of comparator products such as invoice, batch number, expiry date, packaging images, country of purchase, temperature data log, and waybill.

3.1.2 Selection of Reference Products for Combination Products:

Combination products should, in general, be assessed with respect to bioavailability and bioequivalence of individual active substances:

- Either single entity products administered concurrently (in the case of clinically justifiable combinations), or
- Using an existing combination as the reference, which should be an innovator product registered by SAHPRA on safety and efficacy data.

In the former instance, immediate-release oral dosage forms containing a single API may be used as a reference.

NOTE: The use of unregistered old medicines as reference products is not acceptable.

3.2 Biowaivers

In certain instances, the need for a bioequivalence study is waived based on fulfilling the criteria of the so-called BCS-based biowaiver. In such instances, there is no batch used in bioavailability/ bioequivalence study or in clinical testing (biobatch) and by analogy, the batch that is equivalent to a reference product based on satisfactory in vitro dissolution data in at least three different pH media is considered to be the test batch.

The reference product used in the biowaiver in vitro studies should be sourced from the South African market, in case where the reference product is not available in SA refer to section 4.1 For more information, refer to ICH M9.

4. SELECTION OF A REFERENCE PRODUCT FOR COMPARATIVE DISSOLUTION

In vitro dissolution between the biobatch and the reference product used in the bioequivalence study (BE) should be provided in three media (pH 1.2, 4.5 and 6.8) and the quality control (QC) media based on the development of that dissolution method. In addition to this, to ensure interchangeability and substitutability, if a foreign reference product was used in the BE study, comparative *in vitro* dissolution should be performed between the RSA innovator and the foreign reference product used in the BE study in at least the three media (pH 1.2, 4.5 and 6.8) and the quality control (QC) media if applicable.

4.1 Special cases for dissolution requirements

In cases where a BE study is conducted with a foreign reference product and the requirement for comparability with the RSA innovator cannot be met due to the inaccessibility of the RSA innovator, the following mutually exclusive scenarios may apply:

- **Scenario 1:** The RSA innovator is not available on the market, and the reference product used in the BE study and the RSA innovator are from the same marketing authorization holder (MAH).

Where the BE reference product and the RSA innovator product are from the same MAH, a declaration from the Holder of the certificate of registration (HCR) of the RSA innovator product should be provided. The declaration should state that the foreign reference product and RSA reference product are identical in all respects, including formulation, method of manufacture, and manufacturing site.

- **Scenario 2:** The RSA innovator product is not available on the market, and the leading generic product is available on the market.

In a case where the BE study has been conducted with a foreign reference product, and the RSA innovator product is not available for comparative dissolution, the market leader can be used as the reference product, and market leadership should be demonstrated by the applicant consult SAHPRA for guidance.

NOTE: For applications where, comparative dissolution is conducted with market leader rather than the RSA innovator, it is expected that comparative dissolution will not need to be repeated once the SA innovator becomes available on the market.

Scenario 3: The RSA innovator product and the leading generic are not available on the market.

In cases where the RSA innovator product and/ or the leading generic product are not accessible, the authority

should be consulted before submission of an application. Consultation before purchase of the comparator product is strongly recommended. Refer to general information guideline (SAHPGL-HPA-07). Waivers could be accepted on a case-by-case basis. e.g when there is extensive documented use in clinical trials reported in peer-reviewed (accredited) scientific journals, provided that the literature relates to the same formulation, therapeutic indications, and dosages as those proposed for registration. This includes but is not limited to clinical pharmacokinetic (PK) and/ or bioequivalence (BE) data demonstrating therapeutic equivalence to the innovator product and/ the product prequalified by WHO.

5. REQUIREMENTS FOR DEMONSTRATING SIMILARITY

For all applications for generic products, physico-chemical and biopharmaceutics comparison between the reference product and the test product should be performed. Where the reference product is not the RSA innovator, a comparison of the foreign reference product with the RSA innovator is required. In the case of dosage forms not supported by a BE study, at least a comparison of formulation, physico-chemical properties, and impurity profiles depending on the formulation should be demonstrated between test product, RSA innovator and foreign reference. For requirements for bioavailability for alternative dosage forms, refer to the Quality and Bioequivalence guideline (SAHPGL-PEM-02).

6. REGIONAL REQUIREMENTS

Refer to the Quality and Bioequivalence Guideline for the requirements for Module 3.2.R.1, 3.2.R.1.2, 3.2.R.1.3, and 3.2.R.1.4. Please note the following:

- If the biostudy reference product is expired and cannot be used for comparative *in vitro* dissolution studies, a batch of the reference product procured from the same country and manufacturer as the biostudy reference product should be used for dissolution testing. A new biostudy is not required.
- If the biostudy test product (bio-batch) has expired, a representative batch should be selected. The selected representative batch should be the same as the bio-batch, i.e., formulation, specification for the API, the excipients and the FPP manufacturing site, manufacturing process and controls, equipment, and in-vitro performance. Any differences between the bio-batch and the representative batch should be clearly stated and justified. A comparative analysis between the representative batch and the bio-batch should be provided. In addition, the executed Batch Manufacturing records (BMR), multipoint dissolution in the three physiological media at 50 rpm or 75 rpm, as per ICH guideline, and the Certificate of Analysis (CoA) of the representative batch should be provided. Particle size distribution (PSD) characterization of the API lot used in the

manufacture of the representative batch should be provided at d10, d50, and d90, only when biopharmaceutically relevant.

- Motivation for not performing comparative dissolution should be submitted in Module 3.2.R.1.4.

7. REFERENCES

The following related documents are referenced:

- 8.1 For more information see **Appendix II** in the Guideline on the Investigation of Bioequivalence CPMP/EWP/QWP/1401/98 Rev. 1/ Corr**).
- 8.2 ICH M9 Biopharmaceutics Classification System-based Biowaivers.
- 8.3 ICH M13 Bioequivalence for Immediate Release Solid Oral Dosage Forms.
- 8.4 SAHPGL-PEM-02 Quality and Bioequivalence guideline.
- 8.5 SAHPGL-BAU-01 Reliance guideline.
- 8.6 SAHPGL-CEM-01 Clinical guideline.

8. VALIDITY

This guideline is valid for a period of 5 years from the effective date of revision. It will be reviewed on this timeframe or as and when required.

9. APPENDICES

9.1 Appendix I: List of RRAs

- European Union Centralised Procedure (EU CP).
- European Union Decentralised Procedure (EU DCP) (no restrictions on which member state acts as the reference member state).
- Health Canada.
- Medicines and Health Products Regulatory Agency (MHRA (UK)).
- Ministry of Health, Labour and Welfare (MHLW)/ Pharmaceutical and Medical Devices Agency (PMDA) – Japan.
- Swiss Agency for Therapeutic Products (SwissMedic).
- Therapeutic Goods Administration (TGA) -Australia.
- US Food and Drug Administration (US FDA).
- WHO listed Authorities (WLA)- listed for all functions for medicines.

9.2 Appendix II: Decision tree for selection of reference product

